

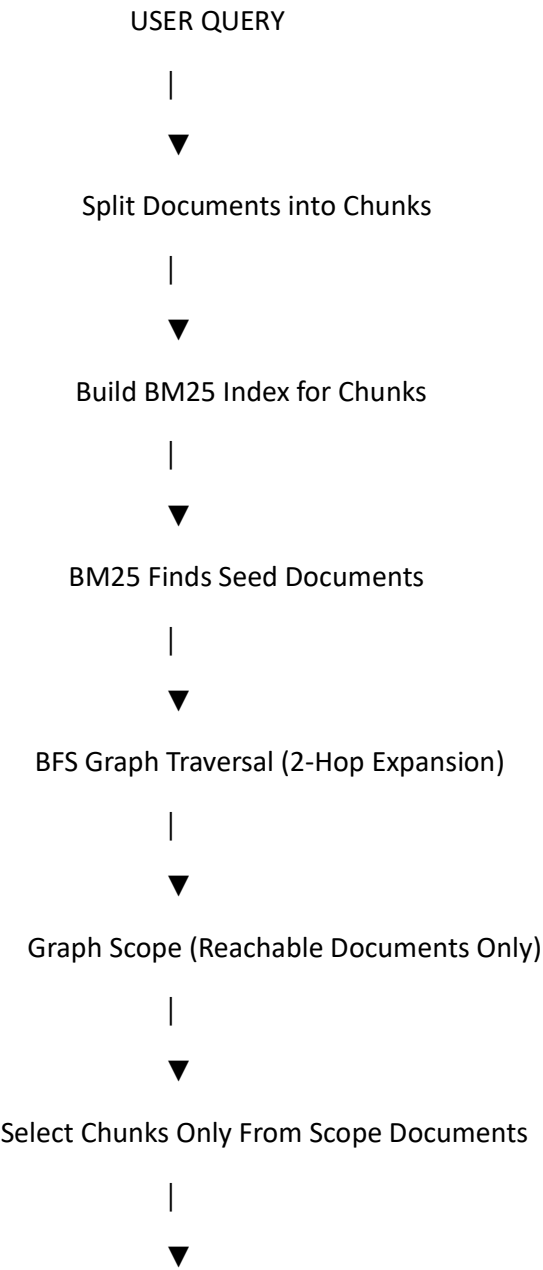
Overall Purpose of Cell 18

The purpose of this cell is:

Retrieve the Top-K most relevant text chunks (instead of full documents) using both BM25 and the Knowledge Graph.

Unlike Cell 16, which ranked **documents**, Cell 18 ranks **chunks** from documents.

Overall Pipeline



BM25 Ranking of Scope Chunks

|



Remove Duplicate / Similar Chunks

|



Select Top-K Chunks (Top 10)

|



Build JSON Payload for Evaluation / LLM

Stage 1 – Chunk Creation

Input:

Knowledge Graph Documents

Example

Document A

Paragraph 1

Paragraph 2

Paragraph 3

Paragraph 4

Paragraph 5

The document is split into overlapping chunks.

Chunk 1

Paragraph 1 + 2

Chunk 2

Paragraph 2 + 3

Chunk 3

Paragraph 3 + 4

Chunk 4

Paragraph 4 + 5

Each chunk becomes a searchable unit.

Stage 2 – Build BM25 Chunk Index

For every chunk

the code calculates

- Tokens
- TF
- DF
- IDF

Result

Chunk Corpus

↓

Chunk1

Chunk2

Chunk3

...

ChunkN

Now BM25 can search chunks.

Stage 3 – Find Seed Documents

The query is compared against
the document title
and document content.

Example

User Query



SQEP Importance

BM25 selects

Top 5 Documents



Doc10

Doc45

Doc71

Doc83

Doc92

These become

Seed Nodes

Notice

the seeds are **documents**, not chunks.

Stage 4 – Graph Traversal

Exactly like Cell 16

Seed Nodes

↓

BFS

↓

Hop 1

↓

Hop 2

Result

Graph Scope

↓

35 Documents

Now

only these

35 documents

are allowed.

Stage 5 – Graph Gate

Suppose

Entire corpus

1000 Documents

Graph Scope

35 Documents

Now

only chunks from

those 35 documents

can participate.

Everything else is ignored.

This is the

Graph Gate

Stage 6 – BM25 Ranking

Now BM25 scores

only chunks

inside

Graph Scope

Each chunk receives

Final Score

=

BM25

+

Hop Bonus

Closer chunks receive
a slightly higher score.

Stage 7 – Remove Duplicates

Many chunks overlap.

Example

Chunk 1

Paragraph 1

Paragraph 2

Paragraph 3

Chunk 2

Paragraph 2

Paragraph 3

Paragraph 4

These are very similar.

The code removes

- Exact duplicates
- Near duplicates

using

- MD5 hash
- Jaccard similarity

Result

Cleaner retrieval.

Stage 8 – Limit Per Document

Suppose

one document has

8 excellent chunks.

Without restriction

the Top 10 could become

Doc1

Doc1

Doc1

Doc1

Doc1

...

Instead

the code says

Maximum

2 Chunks

Per Document

Now

Top 10

contains information

from multiple documents.

Stage 9 – Select Top 10

Finally

Top 10 Chunks

are selected.

Example

1.

Chunk

Doc10

Score

9.2

2.

Chunk

Doc45

Score

8.8

Stage 10 – Build Output

Finally

the retriever creates

```
{
  "predictions": [
    {
      "data": [
        {
          "content": "...",
          "title": "...",
          "reference": "..."
        }
      ]
    }
  ]
}
```

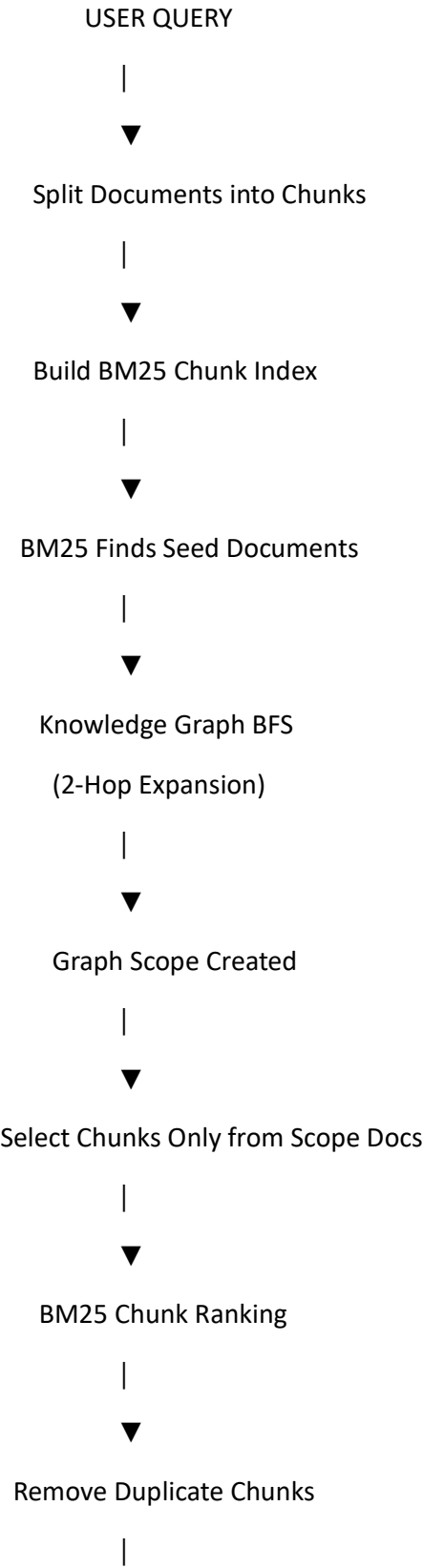
This output is sent

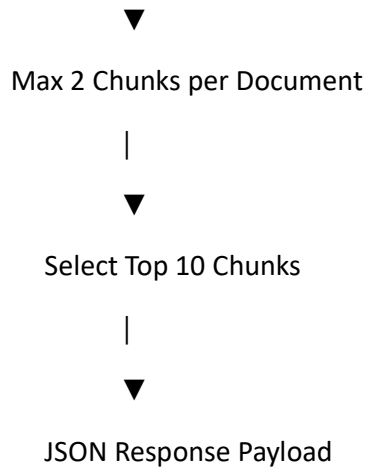
to

- Evaluation
- LLM
- Cell 28

without changing the response format.

Complete Architecture





How it differs from Cell 16

Cell 16

Retrieves **documents**

Ranks full document content

Returns Top-K documents

Builds document context

Better for document-level reasoning

Cell 18

Retrieves **chunks**

Ranks individual text chunks

Returns Top-K chunks

Builds chunk-level context

Better for precise answer extraction

Summary

Cell 18 follows the **same GraphRAG retrieval strategy** as Cell 16:

1. **BM25 identifies the most relevant seed documents.**
2. **BFS expands those seeds through the Knowledge Graph** to create a graph-constrained scope.
3. **Only chunks belonging to documents in that scope are considered** (the graph gate).
4. **BM25 ranks the chunks**, adding a small bonus for chunks from documents closer to the seed nodes.
5. **Duplicate and highly similar chunks are removed**, and the number of chunks per document is limited to improve diversity.
6. The **Top 10 chunks** are returned in a JSON structure ready for downstream evaluation or for building the LLM prompt.

This is a **graph-scoped chunk retriever**, which gives more fine-grained retrieval than Cell 16 while still using the knowledge graph to guide what is searched.

CELL 18 : GRAPH-SCOPED BM25 CHUNK RETRIEVER (CHUNKS)

